

MORRIN AGDM, AB, Canada

WMO: 712570

Lat: **51.6603N**Lon: **112.6750W**Elev: **836**StdP: **91.68**Time Zone: **-7.00 (NAM)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.2	-27.0	-33.7	0.2	-30.2	-30.3	0.2	-26.9	12.0	-4.9	10.8	-7.9	2.6	190	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	29.8	16.9	27.7	16.2	25.7	15.5	18.8	25.9	17.6	25.0	16.5	23.6	4.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.8	21.9	14.8	11.7	20.6	13.7	10.8	19.4	56.9	25.8	52.7	25.2	49.4	23.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.9	8.6	-34.9	33.1	3.1	2.0	-37.1	34.6	-38.9	35.7	-40.6	36.8	-42.8	38.3		
			-34.6	21.0	3.4	1.6	-37.0	22.1	-39.0	23.0	-40.9	23.9	-43.4	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.1	-10.3	-3.7	3.9	10.3	14.2	17.2	15.9	11.0	4.2	-4.4	-10.4	(6)
(7)		DBStd	11.76	8.23	7.61	7.84	5.19	4.30	3.12	2.99	3.44	4.63	5.15	7.25	7.73	(7)
(8)		HDD10.0	3151	622	567	426	188	49	3	0	2	41	189	431	633	(8)
(9)		HDD18.3	5555	880	801	684	432	250	129	58	89	222	438	681	892	(9)
(10)		CDD10.0	685	0	0	1	6	58	129	222	186	72	10	1	0	(10)
(11)		CDD18.3	46	0	0	0	0	1	5	22	15	3	0	0	0	(11)
(12)		CDH23.3	1041	0	0	1	4	54	107	417	332	121	6	0	0	(12)
(13)		CDH26.7	282	0	0	0	0	6	19	125	98	34	0	0	0	(13)
(14)	Wind	WSAvg	3.9	3.7	3.6	4.0	4.6	4.6	4.2	3.5	3.5	3.8	4.0	3.7	3.5	(14)
(15)	Precipitation	PrecAvg	384	14	10	16	26	49	77	65	54	32	16	14	12	(15)
(16)		PrecMax	490	34	24	54	60	93	152	151	108	82	35	48	29	(16)
(17)		PrecMin	229	1	1	3	3	6	33	24	3	5	5	2	1	(17)
(18)		PrecStd	65	8	6	10	14	24	30	30	28	19	9	11	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	6.7	15.2	22.9	27.3	28.8	32.5	32.0	30.6	23.7	14.5	7.5	(19)
(20)			MCWB	2.7	2.8	7.1	10.9	13.1	16.0	17.7	16.9	15.3	12.0	6.9	3.8	(20)
(21)		2%	DB	4.8	3.9	11.4	19.4	24.8	26.2	30.0	29.3	26.7	18.9	10.1	4.4	(21)
(22)			MCWB	1.8	0.9	5.1	8.9	12.2	15.4	18.4	16.2	14.0	9.9	4.6	1.6	(22)
(23)		5%	DB	3.0	1.9	8.5	16.3	22.3	24.1	27.7	27.1	23.8	15.8	7.0	2.3	(23)
(24)			MCWB	0.7	-0.3	3.7	7.2	11.3	14.4	17.5	16.3	13.2	8.4	2.8	-0.1	(24)
(25)		10%	DB	0.7	0.0	5.5	13.4	19.8	21.9	25.5	24.7	20.6	12.8	4.6	0.1	(25)
(26)			MCWB	-1.0	-1.7	2.2	5.7	10.0	13.5	16.7	15.5	11.9	6.8	1.6	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.8	7.9	11.1	14.1	18.3	21.3	19.6	16.0	12.5	7.7	4.0	(27)
(28)			MCDWB	6.3	6.2	14.3	22.1	24.4	25.6	28.0	26.3	27.3	22.5	14.3	7.2	(28)
(29)		2%	WB	1.9	1.2	5.6	9.4	12.8	16.6	19.5	18.1	14.8	10.5	5.1	1.8	(29)
(30)			MCDWB	4.4	3.5	10.5	18.3	22.5	23.9	26.9	25.7	24.9	17.6	9.5	4.3	(30)
(31)		5%	WB	0.8	-0.2	3.7	7.7	11.8	15.4	18.2	17.0	13.5	8.8	3.3	0.0	(31)
(32)			MCDWB	3.0	1.8	8.1	15.3	20.9	22.0	25.6	24.6	22.2	15.1	6.3	2.2	(32)
(33)		10%	WB	-0.9	-1.7	2.2	6.2	10.8	14.2	17.2	15.9	12.3	7.3	1.7	-1.7	(33)
(34)			MCDWB	0.7	0.0	5.2	12.4	18.6	20.5	24.1	23.3	20.2	12.3	4.3	0.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	10.8	12.8	13.6	12.4	14.5	14.3	14.3	13.0	10.6	9.4	(35)
(36)			MCDBR	11.2	11.5	14.3	18.7	17.9	16.3	18.2	19.0	20.0	17.7	13.7	10.8	(36)
(37)			MCWBR	8.9	9.1	8.8	9.7	8.2	7.4	8.3	8.1	9.0	9.7	9.0	8.5	(37)
(38)		5% WB	MCDWR	10.8	11.3	13.7	17.4	16.1	14.0	16.2	16.1	18.5	16.8	12.8	10.7	(38)
(39)			MCWBR	8.6	9.1	8.8	9.3	7.5	7.2	8.2	7.7	8.7	9.4	8.8	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.266	0.272	0.284	0.338	0.363	0.359	0.356	0.373	0.323	0.283	0.254	0.243	(40)	
(41)		taud	2.329	2.404	2.460	2.327	2.310	2.363	2.398	2.333	2.442	2.535	2.484	2.371	(41)	
(42)		Ebn at Noon	757	869	926	900	889	894	891	854	866	850	788	737	(42)	
(43)		Edh at Noon	61	76	87	113	121	116	110	112	89	66	52	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.09	2.12	3.51	4.74	5.74	6.14	6.53	5.29	3.72	2.24	1.19	0.85	(44)	
(45)		RadStd	0.08	0.17	0.29	0.38	0.38	0.37	0.33	0.37	0.39	0.26	0.10	0.08	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page